

VAIBHAV VIKAS RANJAN

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Education

International Institute of Information Technology, IIIT-NR

B.Tech Data Science and Artificial Intelligence

Nov 2022 - July 2026

Naya Raipur, Chhattisgarh

Experience

Accure Inc.

Jan 2026 – July 2026

AI Engineer Intern

Bangalore, India

- Improved **LLM reasoning reliability** for an enterprise **generative AI** platform (AccureIQx), evaluated through a **seven-dimension evaluation framework** and **six-criterion validation rubric**, by designing and governing **agentic AI workflows**, **RAG pipelines**, and **prompt-engineering** pipelines for **enterprise search** and reasoning use cases.
- Strengthened adversarial robustness of **agentic reasoning** systems, achieving high **anomaly-detection accuracy** across **five stress-test scenarios**, by engineering synthetic **cybersecurity** incident datasets with realistic log data.
- Cut manual weekly reporting time by **35%**, saving analyst hours, by building automated **Python (Pandas, NumPy)** analytics pipelines integrating multi-source **EDA** data via **REST APIs**.

Taskar – India's 1st Healthcare Mall

May 2025 – July 2025

Software Developer Intern

Remote, India

- Reduced manual reporting workflows by **40%**, cutting process time, by building **microservice-based RESTful API** integrations to retrieve and analyze live **healthcare and inventory data** for business stakeholders.
- Improved **analytics dashboard** load times by **30%**, per page-render benchmarks, by optimizing **UI rendering** and **asynchronous data fetching** with **React Hooks**.
- Increased user engagement by **15%** and cross-team adoption, confirmed by **A/B test** results, by redesigning dashboard layouts based on internal stakeholder feedback.

Research Designs and Standards Organisation (R.D.S.O)

May 2024 – July 2024

Software Engineering Intern

Lucknow, India

- Supported **500+ concurrent users** at **99.9% uptime**, verified through system load testing, by engineering a responsive, **service-oriented attendance tracking system (HTML, CSS, JavaScript, PHP, MySQL)**.
- Cut login times by **60%** for **1,000+ daily users**, based on authentication response benchmarks, by designing an optimized **authentication workflow** and adding **AJAX-based** real-time interactivity that boosted retention **20%**.
- Validated production-grade performance under high concurrency, using **Apache JMeter** load and scalability tests, by conducting systematic **stress testing** across peak-usage scenarios.

Projects

CFR Compliance Checker: Agentic Contract Auditor | *Python, FastMCP, Agno, FastAPI, PostgreSQL* | [Github](#)

- Architected an **agentic compliance pipeline** (security → **eCFR retrieval** → deterministic rules → **LLM reasoning** → **verification** → human review) with **evidence provenance** attached only by the retrieval layer, routing unresolved cases to **NEEDS REVIEW** instead of silent acceptance.
- Built a **human-in-the-loop review** lifecycle with **optimistic concurrency** and an immutable audit trail, plus an optional **PostgreSQL** backend with versioned migrations, validated by **268 offline tests**.

Incident Response Engine | *Python, Agno, FastMCP, Kafka, ChromaDB, Scikit-learn* | [Github](#)

- Engineered a **Kafka** event pipeline gating an LLM agent pipeline behind a deterministic z-score filter and **ML anomaly gate (IsolationForest + LOF)**, live-validated end-to-end against a real broker and sandboxed **MCP** diagnostics.
- Designed a **two-phase Investigator Agent** separating tool-calling from analysis to prevent self-reported tool-use, and enforced a hard **human-approval gate (Slack + dashboard HITL)** before any remediation, validated by a **220-test suite** and a live tool-calling run.

Multi-Agent Orchestrator: GitHub Issue to PR Pipeline | *Python, LangGraph, Docker, GitHub API* | [Github](#)

- Architected a stateful **LangGraph** pipeline (Researcher, Planner, Coder, TestWriter, Verifier, PRWriter) with conditional **retry/escalation routing** and **Docker-sandboxed verification**, gating every PR behind mandatory **human approval** before publication.
- Validated the real end-to-end path with a mocked integration suite against a stateful fake GitHub plus a credential-gated **live suite** (real GitHub, LLM, Docker) confirming no PR is ever created without explicit approval.

Technical Skills and Interests

Languages: Python, SQL, C++, JavaScript, HTML/CSS

ML/DL Frameworks: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV

LLM/Agentic Stack: LangChain, LangGraph, Agno, MCP, RAG, Chainlit, Hugging Face

Backend & Infrastructure: FastAPI, Streamlit, PostgreSQL, ChromaDB, Docker, AWS, Kafka

Tools: Git, Tableau

AI/ML Domains: Generative AI, LLMs, RAG (Retrieval-Augmented Generation), Agentic AI, NLP, Machine Learning, Deep Learning, Computer Vision, Anomaly Detection, Prompt Engineering, Model Evaluation, Data Engineering, Data Analysis, Full Stack Development